Islamic University of Technology

ARTIFICIAL INTELLIGENCE

CSE 4617

Model-Free Policy Evaluation

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Contents

1 Introduction and Motivation

Policy evaluation is the task of estimating the expected discounted return of a target policy π_e using data that was collected by some other behavior policy π_b. When the environment dynamics P and reward function R are unknown, *model-free* estimators operate directly on logged transitions without assuming an explicit model. Robust off-policy evaluation (OPE) is particularly important in domains such as healthcare, recommender systems, and robotics, where online testing of new policies can be risky or expensive [1, 2]. It is also widely used for hyperparameter tuning and automated policy selection without requiring costly real-world trials.

Despite its importance, model-free policy evaluation is often presented only in mathematical form. Before formalizing the setup, there's a brief intuitive overview of what model-free evaluation means, how it contrasts with model-based approaches, and why it is attractive in practice.

2 What is Model-Free Policy Evaluation?

Policy evaluation aims to estimate the long-term return of a target policy π_e without executing it in the environment. Formally, the objective is to estimate

$$J(\pi_e) = \mathbb{E}_{\pi_e}\Big[\sum_{t=0}^{\infty} \gamma^t r_t\Big],$$

the expected discounted return of π_e.

Model-based vs. Model-free

Two main approaches exist:

- **Model-based:** First learn or assume an explicit model of dynamics P and rewards R, then simulate rollouts of π_e in this model to estimate $J(\pi_e)$.
- **Model-free:** Skip building a model. Instead, directly use logged experience tuples (s, a, r, s') to estimate the return of π_e. This avoids model misspecification but requires statistical corrections to handle the fact that data usually comes from a different policy π_b.

Why model-free?

In many settings (e.g., healthcare, recommendation systems, robotics), building an accurate model of the environment is infeasible or unsafe. Model-free methods leverage existing data directly, making them more practical when a trustworthy simulator cannot be assumed.

This motivates the following sections, where different families of model-free OPE methods are reviewed.

3 Preliminaries

An MDP is defined as $\mathcal{M} = (\mathcal{S}, \mathcal{A}, P, R, \gamma)$ where:

- $\mathcal{S}$: state space,
- $\mathcal{A}$: action space,
- $P(s' \mid s, a)$: transition dynamics,
- $R(s, a)$: reward distribution,
- $\gamma \in [0, 1)$: discount factor.

A policy π maps states to action distributions. For a trajectory $\tau = (s_0, a_0, r_0, s_1, a_1, r_1, \dots)$, the (discounted) return is

$$G(\tau) = \sum_{t=0}^{\infty} \gamma^t r_t.$$

The value of π_e is

$$V^{\pi_e} = \mathbb{E}_{\tau \sim (\pi_e, P)}\big[G(\tau)\big].$$

In off-policy evaluation, data are generated by a different policy π_b. The dataset is

$$\mathcal{D} = \{(s_i, a_i, r_i, s'_i)\}_{i=1}^{n},$$

and the goal is to estimate V^{π_e} from $\mathcal{D}$ alone, without access to P or R.

4 Taxonomy of Model-Free OPE Methods

The key families of model-free estimators, each with core equations:

4.1 Sample-Based Evaluation (Monte Carlo)

The simplest method estimates the value of a policy by averaging returns from on-policy trajectories:

$$\widehat{J}_{\text{MC}} = \frac{1}{N} \sum_{i=1}^{N} G^{(i)}, \qquad G^{(i)} = \sum_{t=0}^{T_i - 1} \gamma^t r_t^{(i)}.$$

This estimator is unbiased if data come from π_e, but impractical in offline OPE where only off-policy data are available.

4.2 Importance Sampling (IS)

IS reweights trajectories from π_b to mimic π_e. Trajectory-level IS:

$$\widehat{J}_{\text{IS}} = \frac{1}{N} \sum_{i=1}^{N} \Big(\prod_{t=0}^{T_i - 1} \rho_t^{(i)} \Big) G^{(i)},$$

with $\rho_t^{(i)}$ denoting importance ratios. Per-decision IS (PDIS) applies weights incrementally:

$$\widehat{J}_{\text{PDIS}} = \frac{1}{N} \sum_{i=1}^{N} \sum_{t=0}^{T_i - 1} \gamma^t \Big(\prod_{k=0}^{t} \rho_k^{(i)} \Big) r_t^{(i)}.$$

Self-normalized IS reduces variance at the cost of slight bias. **Diagnostics:** Effective sample size (ESS) and weight histograms are essential; very low ESS signals unreliable estimates [3].

4.3 Regression / Fitted Q Evaluation (FQE)

FQE fits Q_θ via Bellman regression:

$$y_i^{(k)} = r_i + \gamma \, \mathbb{E}_{a' \sim \pi_e(\cdot | s_i')} \big[Q_k(s_i', a') \big].$$

Then update $Q_{k+1} \leftarrow \arg\min_\theta \sum_i (Q_\theta(s_i, a_i) - y_i^{(k)})^2$. FQE has lower variance than IS but may be biased if function approximation is poor [4, 5].

4.4 Doubly Robust (DR) Estimators

DR combines regression baselines with IS residuals:

$$\widehat{J}_{\mathrm{DR}} = \frac{1}{N} \sum_{i=1}^{N} \left[\hat{V}(s_0^{(i)}) + \sum_{t=0}^{T_i-1} \gamma^t \Big(\prod_{k=0}^{t} \rho_k^{(i)} \Big) (r_t^{(i)} + \gamma \hat{V}(s_{t+1}^{(i)}) - \hat{Q}(s_t^{(i)}, a_t^{(i)})) \right].$$

DR is consistent if either the model or weights are correct, making it robust in practice [6].

4.5 Density-Ratio / Distribution Correction (DualDICE)

These methods estimate occupancy ratios $w(s,a) = \frac{d^{\pi_e}(s,a)}{d^{\pi_b}(s,a)}$, then compute expectations under π_e using w. They are behavior-agnostic but require solving difficult optimization problems [7].

4.6 Model-Free Monte Carlo–like (MFMC)

MFMC stitches together one-step transitions into pseudo-trajectories using similarity matching. It works under smoothness assumptions on dynamics and is useful when only one-step logs are available [8].

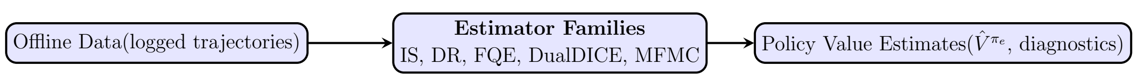

Figure 1: Pipeline of model-free off-policy policy evaluation.

5 Theoretical Perspectives

- **Bias–variance tradeoff:** IS is unbiased but high-variance; FQE reduces variance but may be biased; DR balances both.
- **Asymptotics & inference:** Recent work develops inference frameworks (e.g., Z-estimation, bootstrap CIs) for FQE-like methods under mild conditions [5].
- **Support / coverage:** Reliable OPE requires overlap between π_e and π_b; without it, estimates rely on extrapolation.

6 Recommended Experimental Protocol

6.1 Environments and Datasets

1. **Tabular gridworlds:** useful for ground-truth analysis.
2. **Classic control:** e.g., CartPole, to study short horizons.
3. **Continuous control:** e.g., D4RL tasks such as HalfCheetah-Medium and Hopper-Medium-Expert.

6.2 Behavioral Data Generation

- Generate datasets with varying policy shifts: near on-policy, moderate shift, and extreme shift (low overlap).
- Use multiple random seeds for reproducibility.
- Record or estimate π_b, required for IS-based methods.

6.3 Baselines and Metrics

- **Methods:** IS, SNIS, PDIS, FQE, DR, DualDICE, MFMC.
- **Metrics:** RMSE vs ground truth (if available), bias, variance, CI coverage, ESS.
- **Reporting:** mean $\pm$ std across seeds; scatter plots of ESS vs error; CI coverage tables.

6.4 Hyperparameters & Tuning

- Report learning rates, architectures, regularization, early stopping.
- Use cross-validation when possible; for FQE, use validation transitions for stopping.

7 Comparison Table

8 Open Challenges and Future Directions

- Scaling to high-dimensional states and long horizons while controlling variance.
- Handling partial observability and hidden confounding.

Method	Bias	Variance	Practical notes
MC (sample-based)	Low (if on-policy)	Moderate–high	Requires on-policy rollouts; impractical in offline OPE
IS / PDIS	Low (unbiased)	High	Use ESS; fails for long horizon
FQE (regression)	Model bias possible	Moderate	Stable with coverage; needs early stopping
Doubly Robust	Small (robust)	Moderate	Requires good baseline or weights
DualDICE / density-ratio	Depends	Moderate	Optimization-heavy; behavior-agnostic
MFMC	Approximate	Low–Moderate	Needs smooth dynamics; good for one-step logs

Table 1: Concise method comparison for quick reference.

- Creating benchmarks and datasets that better capture real-world distribution shifts.
- Multi-policy evaluation and shared-sample approaches for sample efficiency (e.g., CAESAR) [9].

9 Research Papers

9.1 Model-Free Monte Carlo-like Policy Evaluation

Fonteneau et al. introduced the *Model-Free Monte Carlo-like (MFMC)* method [8], which aims to evaluate policies from datasets consisting only of one-step transitions rather than full trajectories.

9.1.1 Core Ideas

The central idea is to construct artificial trajectories by 'stitching" together transitions from the dataset using nearest-neighbor matching in state space. These pseudo-trajectories are then used to compute approximate cumulative returns, much like Monte Carlo evaluation would with genuine episodes.

9.1.2 Problems Addressed

Traditional Monte Carlo evaluation requires full episodes collected under the policy of interest, which may not be feasible in offline or costly environments. The MFMC method addresses this by working directly with logged transition data, without needing to interact with the environment.

9.1.3 Solutions and Methodology

MFMC introduces a matching strategy based on state similarity to link one-step transitions into coherent trajectories. Under Lipschitz continuity assumptions for the dynamics and rewards, the authors derive theoretical error bounds for the MFMC estimates.

9.1.4 New Perspectives and Impact

This work provided one of the earliest systematic approaches to offline policy evaluation without requiring model knowledge. It offered a statistical learning perspective, connecting RL with regression-based error analysis.

9.1.5 Use Cases

Potential applications include:

- **Healthcare:** Evaluating treatment strategies from observational patient data without running new trials.
- **Finance:** Assessing trading strategies from logged historical market transitions.
- **Robotics:** Testing control policies when running real-world experiments is expensive or risky.

9.1.6 Recent Developments

The MFMC approach inspired later offline RL methods, including density-ratio based estimators and robust OPE techniques. Its emphasis on error bounds and data reuse continues to influence modern OPE theory and practice.

9.2 Doubly Robust Off-Policy Value Evaluation

Jiang and Li introduced the *Doubly Robust (DR)* estimator for reinforcement learning [6], extending earlier DR methods from contextual bandits to sequential decision-making. Their approach addresses key limitations of existing off-policy evaluation techniques.

9.2.1 Core Ideas

The DR estimator combines the strengths of regression-based and importance-sampling-based estimators. It leverages approximate value function estimates to reduce variance, while preserving the unbiasedness properties of importance sampling when either the model or the weights are correct.

9.2.2 Problems Addressed

Traditional regression-based approaches suffer from bias due to model misspecification, while importance sampling suffers from prohibitively high variance, especially in long-horizon settings. The DR method was designed to achieve low variance while remaining unbiased, thereby offering robustness against errors in either component.

9.2.3 Solutions and Methodology

The authors derive a recursive DR formulation that uses value function approximations to control variance. They prove that when the approximator is accurate, the DR variance approaches the theoretical Cramér–Rao lower bound. Additionally, they propose extensions such as k-fold DR to make more efficient use of data, and DR-v2 for variance reduction under strong assumptions.

9.2.4 New Perspectives and Impact

This work positioned DR as a 'best-of-both-worlds" estimator in reinforcement learning. It not only influenced later developments in safe policy improvement and offline RL but also strengthened the bridge between causal inference and reinforcement learning by showing how counterfactual reasoning can be robustly combined with function approximation.

9.2.5 Use Cases

The DR estimator is particularly well-suited for domains where safety and reliability are critical:

- **Healthcare:** Evaluating new treatment strategies from logged data without introducing risk to patients.
- **Robotics:** Safely assessing policies before deployment on physical systems to avoid costly or dangerous errors.
- **Business and Marketing:** Testing new recommendation or advertising strategies using historical interaction logs.

9.2.6 Recent Findings

Experiments in benchmark domains such as Mountain Car, Sailing, and real-world datasets demonstrated that DR consistently outperforms both IS and regression-based methods. In safe policy improvement tasks, DR enabled more aggressive acceptance of beneficial policies while maintaining safety guarantees. This methodology has since been extended into advanced DR variants and integrated into modern offline RL frameworks.

9.3 CAESAR — Multiple-Policy Evaluation via Density Estimation

Chen, Pacchiano, and Paschalidis introduce CAESAR, which addresses the multi-policy evaluation problem by estimating coarse visitation distributions for many policies and deriving an optimal shared sampling distribution that enables simultaneous evaluation of many target policies with improved sample efficiency [9].

9.3.1 Core Ideas

CAESAR reduces redundant sampling across K target policies by (1) estimating coarse visitation distributions for each policy, and (2) solving an optimization to obtain a shared sampling distribution that supports accurate importance-weighted evaluation for all policies.

9.3.2 Problems Addressed

Naive evaluation scales poorly with K (linear or worse), and separate evaluation runs waste samples. CAESAR targets an instance-dependent sample complexity that can scale sublinearly in K under favorable overlap conditions.

9.3.3 Methodology and Solutions

The method runs a lightweight estimation phase per policy to get coarse occupancies, then optimizes a sampling distribution (DualDICE-inspired) to minimize a worst-case variance criterion across all target policies. The paper provides non-asymptotic instance-dependent bounds and empirical validation.

9.3.4 Use Cases and Impact

Useful for hyperparameter search, policy selection, automated ML pipelines with many candidate policies, and large-scale recommender-system evaluations. The OpenReview page and paper are available online [9].

Synthesis. Taken together, these works trace the trajectory of model-free policy evaluation from early importance sampling and eligibility-trace approaches toward more statistically principled and data-efficient estimators such as doubly robust methods, fitted value-based techniques, and distribution-correction frameworks like DualDICE. More recent contributions, such as MFMC and multi-policy evaluation strategies, emphasize practical applicability and scalability. Viewed as a whole, the literature reflects a steady evolution: from variance-heavy but conceptually simple estimators, to hybrid and robust methods balancing bias and variance, and finally to approaches designed for deployment at scale and under structural constraints. This progression underscores both the maturity of the field and the open challenges that remain in bridging theoretical guarantees with practical reliability.

Insights

Model-free off-policy evaluation occupies a critical intersection between theoretical rigor and practical applicability. While in principle the objective is to design estimators that are unbiased and low in variance, real-world applications often involve noisy logged data, model misspecification, and insufficient support. Based on this perspective, several key considerations emerge:

- **Prioritize diagnostics before estimator selection.** Assess support overlap, examine weight distributions, and compute effective sample size (ESS) prior to committing to a particular estimator. Without such diagnostics, even sophisticated estimators are unlikely to yield reliable results.
- **Emphasize robustness over theoretical optimality.** In applied settings, doubly robust approaches and conservative reporting practices often outperform theoretically elegant but fragile methods, particularly under distributional shift or when function approximation is required.
- **Adopt a multi-policy perspective.** Practical deployments typically involve comparing a collection of candidate policies rather than evaluating a single one. Approaches that improve sample efficiency across multiple target policies can therefore provide greater practical utility.

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